

PAPER

nf-core/viralmetagenome: A Novel Pipeline for Untargeted Viral Genome Reconstruction

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Abstract

Motivation: Eukaryotic viruses present significant challenges for genome reconstruction and variant analysis due to their extensive diversity and potential genome segmentation. While de novo assembly followed by reference database matching and scaffolding is common, manual execution is labor-intensive, particularly when extensive reference curation is required. We address the critical need for an automated, scalable pipeline for efficient viral metagenomic analysis without manual intervention.

Results: We present nf-core/viralmetagenome, a comprehensive viral metagenomic pipeline for untargeted genome reconstruction and variant analysis of eukaryotic DNA and RNA viruses. Viralmetagenome is implemented as a Nextflow workflow that processes short-read metagenomic samples to automatically detect and assemble viral genomes, while also performing variant analysis. The pipeline features automated reference selection, consensus quality control metrics, comprehensive documentation, and seamless integration with containerization technologies, including Docker and Singularity. We demonstrate the utility and accuracy of our approach through validation on both simulated and real datasets, showing robust performance across diverse viral families in metagenomic samples.

Availability: nf-core/viralmetagenome is freely available at <https://github.com/nf-core/viralmetagenome> with comprehensive documentation at <https://nf-co.re/viralmetagenome>

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Supplementary information: Supplementary data are available at <https://github.com/Joon-Klaps/nf-core-viralmetagenome-manuscript> online.

Key words: viralmetagenome, bioinformatic pipeline, nextflow, viral metagenomics, viral assembly, viral variant analysis

Introduction

Reconstructing viral genomes from metagenomic sequencing data presents considerable computational challenges, particularly for viruses exhibiting extensive genetic diversity (Baaijens et al., Meleshko et al.). This challenge is amplified in segmented viruses such as influenza, rotavirus, and bunyaviruses, where individual segments can evolve under distinct selective pressures and reassortment complicates reconstruction. While some pipelines target specific viruses and their subtypes (Shepard et al.), accurate genome reconstruction of samples with unknown references typically requires time-consuming manual curation of contigs and reference matching (de Vries et al.). This makes large-scale metagenome studies or rapid outbreak responses impractical.

To address these limitations, we developed nf-core/viralmetagenome, a comprehensive pipeline specifically designed for untargeted

viral genome reconstruction. The pipeline is developed using Nextflow (Di Tommaso et al.) within the nf-core framework (Ewels et al. b, Langer et al.), ensuring reproducibility through containerization with Docker (Merkel) and Singularity (Kurtzer et al.), and enabling portability across computational platforms such as local desktops, high-performance clusters and cloud environments.

Pipeline Description

nf-core/viralmetagenome implements an automated workflow for *de novo* assembly, reference matching, and iterative consensus refinement to reconstruct viral genomes without prior target knowledge. The pipeline consists of five major stages: read preprocessing, metagenomic diversity assessment, contig assembly and scaffolding, iterative consensus refinement with variant analysis, and quality control (Figure 1). Unless

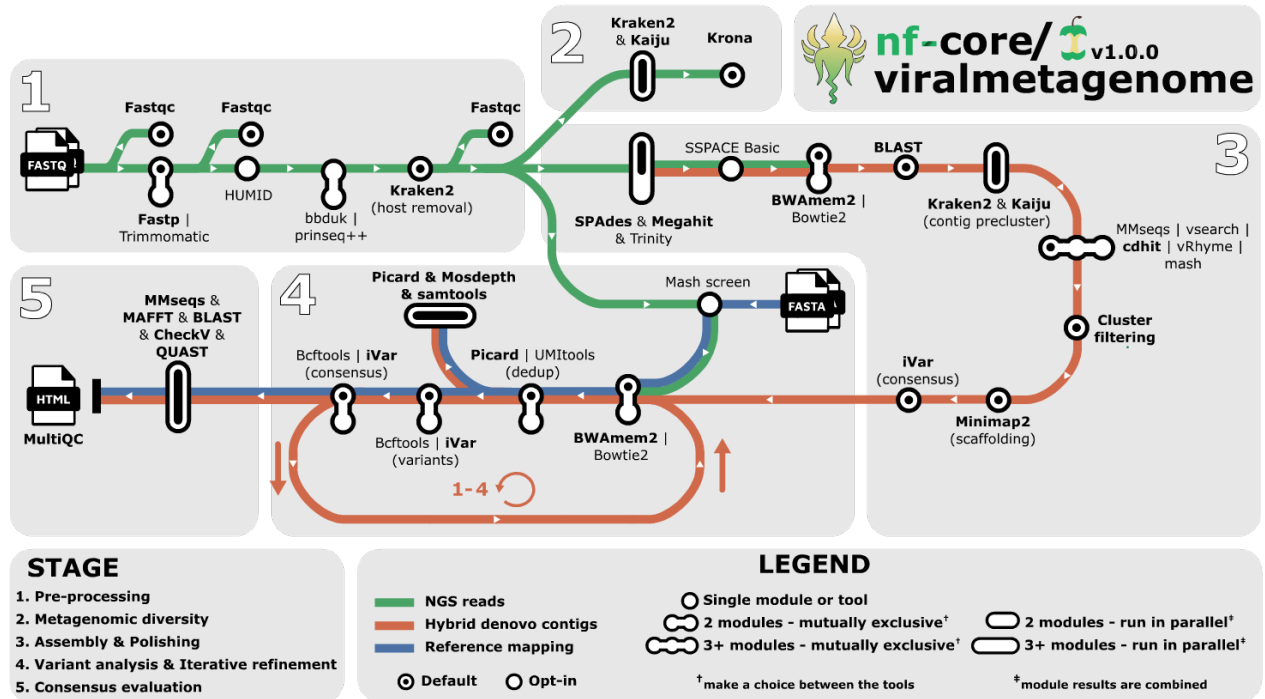


Fig. 1. Overview of the nf-core/viralmetagenome pipeline for untargeted viral genome reconstruction. nf-core/viralmetagenome processes short-read data through pre-processing, metagenomic diversity assessment, de novo assembly with multiple assemblers, scaffolding with automated reference identification and taxonomy-guided clustering of contigs, and iterative consensus refinement through read mapping and variant calling. Quality control metrics, assembly statistics, and coverage data are integrated into interactive MultiQC reports and standardised overview tables for downstream analysis

otherwise specified, the pipeline offers multiple options to accommodate established user workflows and preferences. Detailed tool descriptions are provided in Supplementary Table 1.

Read preprocessing

Input reads listed in sample sheets containing sample names and short-read FASTQ paths are preprocessed using FastQC and trimmed for adapters using fastp (Chen et al.) (default) or Trimmomatic (Bolger et al.). Fastp offers automated adapter detection and faster processing (Chen et al.). UMI deduplication is implemented using HUMID (Laros) for raw reads, or UMI-tools after mapping (Smith et al.). Multiple sequencing runs can be merged after trimming by specifying merge group identifiers in the sample sheet. Filtering with bbduk (Bushnell) or PRINSEQ++ (Cantu et al.) removes low-complexity sequences, while host reads are removed with Kraken2 (Wood et al.).

Metagenomic diversity assessment

Taxonomic classification of preprocessed reads is performed using two complementary approaches - Kaiju (Menzel et al.) and Kraken2 (Wood et al.) - to maximise detection sensitivity across diverse viral families. Results from both classifiers are visualised using Krona (Ondov et al. a).

De novo assembly and clustering

The assembly workflow employs multi-assembler approaches followed by clustering and scaffolding of contigs independent

of the original assembler. *De novo* assembly uses SPAdes (Meleshko et al.) (RNAviral mode), MEGAHIT (Li et al.), and Trinity (Grabherr et al.), capitalizing on distinct algorithms to maximise genome recovery across diverse viral families and read depths. Reference identification uses BLASTn (Altschul et al.) against the Reference Viral Database (RVDB) (Goodacre et al.), retaining the top five hits to identify related genomic segments for scaffolding. Clustering is performed in two sequential stages. First, taxonomic pre-clustering groups contigs based on taxonomic classification using Kraken2 (Wood et al.) and Kaiju (Menzel et al.), with optional filtering to focus on specific taxonomic clades for more targeted analyses. Second, nucleotide similarity clustering within taxonomic groups is performed using CD-HIT-EST (Li and Godzik) (default), VSEARCH (Rognes et al.), MMseqs2 (Steinegger and Söding), vRhyme (Kieft et al. a), or Mash (Ondov et al. b). All tools are valid options, though performance may vary depending on the dataset (Zielezinski et al., Steinegger and Söding). As an optional filtering step of contig clusters following assembly and extension, reads can be mapped to all contigs using BWA-MEM2 (Vasimuddin et al.) (default), or Bowtie2 (Langmead et al.). Clusters are filtered based on the total percentage of reads mapping to all contigs within a cluster, allowing identification and removal of clusters that likely represent assembly artefacts resulting from low read coverage. For the final scaffolding step, all cluster members are mapped to the cluster representative or centroid using Minimap2 (Li), followed by consensus calling with iVar (Grubaugh et al.) to generate reference-assisted assemblies.

Iterative consensus refinement and variant calling

The consensus module supports external reference-based analysis and scaffold refinement. Users can provide specific reference genomes; if a set is provided, Mash (Ondov et al. b) selects the most similar one. The selected reference genome undergoes a single round of refinement to generate the consensus. In contrast, scaffold refinement employs an iterative approach, performing up to 4 iterative cycles (default 2) to polish the consensus sequence. Both processes map reads using BWA-MEM2 (Vasimuddin et al.), or Bowtie2 (Langmead et al.), followed by variant calling with BCFtools (Danecek et al.) or iVar (Grubaugh et al.). Benchmarking (Bassano et al.) indicated that BCFtools outperforms iVar in precision and recall, while iVar identified more low-frequency variants. Users can prioritise sensitivity or specificity when selecting the variant caller. Final variant calling is performed on the polished consensus to characterise single nucleotide variants.

Consensus Quality Control

Quality control employs CheckV (Nayfach et al.) to assess completeness, BLASTn (Altschul et al.) for reference similarity, and MMseqs2 (Steinegger and Söding) against the annotated database Virosaurus (Gleizes et al.). These analyses enable species identification, genomic segment classification, host prediction, and any other additional metadata embedded within the reference databases. The refinement progression is evaluated through sequence alignment with MAFFT (Katoh et al.), which compares final consensus genomes against *de novo* contigs, intermediate consensus sequences from iterative cycles, and the scaffolding reference. All tool metrics are compiled into an interactive MultiQC report (Ewels et al. a). Additionally, key metrics are extracted from the MultiQC report and compiled into standalone overview tables to facilitate downstream analysis across all processed samples.

Applications

We validated performance on real metagenomic samples spanning human and plant pathogens. nf-core/viralmetagenome generated high-quality or near-complete genomes across viral families including segmented (Lassa virus, Orthornairovirus, Tomato spotted wilt tospovirus) and non-segmented viruses (SARS-CoV-2, West Nile virus, Potato virus Y, Youcai mosaic virus, and Monkeypox virus; Supplementary methods S1). Processing these 28 samples required 412 CPU hours and a maximum of 79GB RAM on a high-performance cluster. For local desktop use, we suggest to use the `-profile local` option which disables the memory intensive Kaiju database in favor of a smaller one (viral refseq), reducing maximum RAM usage to approximately 18GB.

To assess performance under challenging scenarios, we conducted two targeted validation experiments. First, we evaluated the reconstruction of multiple viral genomes in a simulated mixture sample (Supplementary Methods S2). The pipeline successfully separated coinfecting strains when the genetic divergence between them was reasonably high ($\leq 88.7\%$ ANI; Supplementary Figure 1). Second, we investigated the influence of the scaffolding reference using real Lassa virus sequencing data (Supplementary Methods S3). We first established a baseline consensus reconstruction by providing the unclustered-RVDB as the reference input, and then compared this baseline against consensus reconstructions

generated using single, increasingly dissimilar scaffolding references (ranging from 64% to 100% ANI).

Our results showed that sequencing depth dictates the importance of the scaffolding reference. For the high-coverage S segments, *de novo* assembly alone was often sufficient to reconstruct the full genome, so the choice of scaffolding reference had minimal impact (Figure 2A-B). In contrast, for the lower-coverage L segments, *de novo* assembly often resulted in fragmented contigs. Here, the use of a closely related external reference was critical, allowing the pipeline to bridge gaps and produce a significantly more complete and accurate consensus genome compared to using *de novo* contigs alone (Figure 2C-D).

These findings emphasise the need to keep databases like RVDB (Goodacre et al.) up-to-date. Since nf-core/viralmetagenome is primarily designed for eukaryotic viruses, users interested in bacteriophage analysis are encouraged to explore pipelines designed to target phages such as VIRify (Rangel-Pineros et al.), VIBRANT (Kieft et al. b), VirSorter2 (Guo et al.).

Conclusion

nf-core/viralmetagenome addresses a critical need in viral genomics by providing an automated, scalable solution for untargeted viral genome reconstruction. The pipeline successfully automates the traditionally time-consuming and manual execution process of viral genome reconstruction from short-read metagenomic data through its integrated workflow of *de novo* contig assembly, automated reference selection, clustering algorithms, and iterative refinement strategies. Our validation results indicate that the pipeline is suitable for a wide range of eukaryotic viral families, it removes the burden of extensive manual curation, and enables consistent, high-quality genome reconstruction with reproducible results across varied computational settings. As viral surveillance and outbreak response increasingly rely on metagenomic sequencing, automated pipelines like nf-core/viralmetagenome will be essential for the timely identification of pathogen strains. The pipeline represents a significant step forward in making viral genome reconstruction accessible to researchers without requiring extensive bioinformatics expertise, facilitating broader adoption of metagenomic approaches in viral research and public health applications.

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Author Contributions

J.K. designed and implemented the pipeline, performed validation analyses, and wrote the manuscript. P.L. and L. E. K. supervised the project and provided critical feedback. The nf-core community contributed to maintaining the pipeline. All authors reviewed and approved the final manuscript.

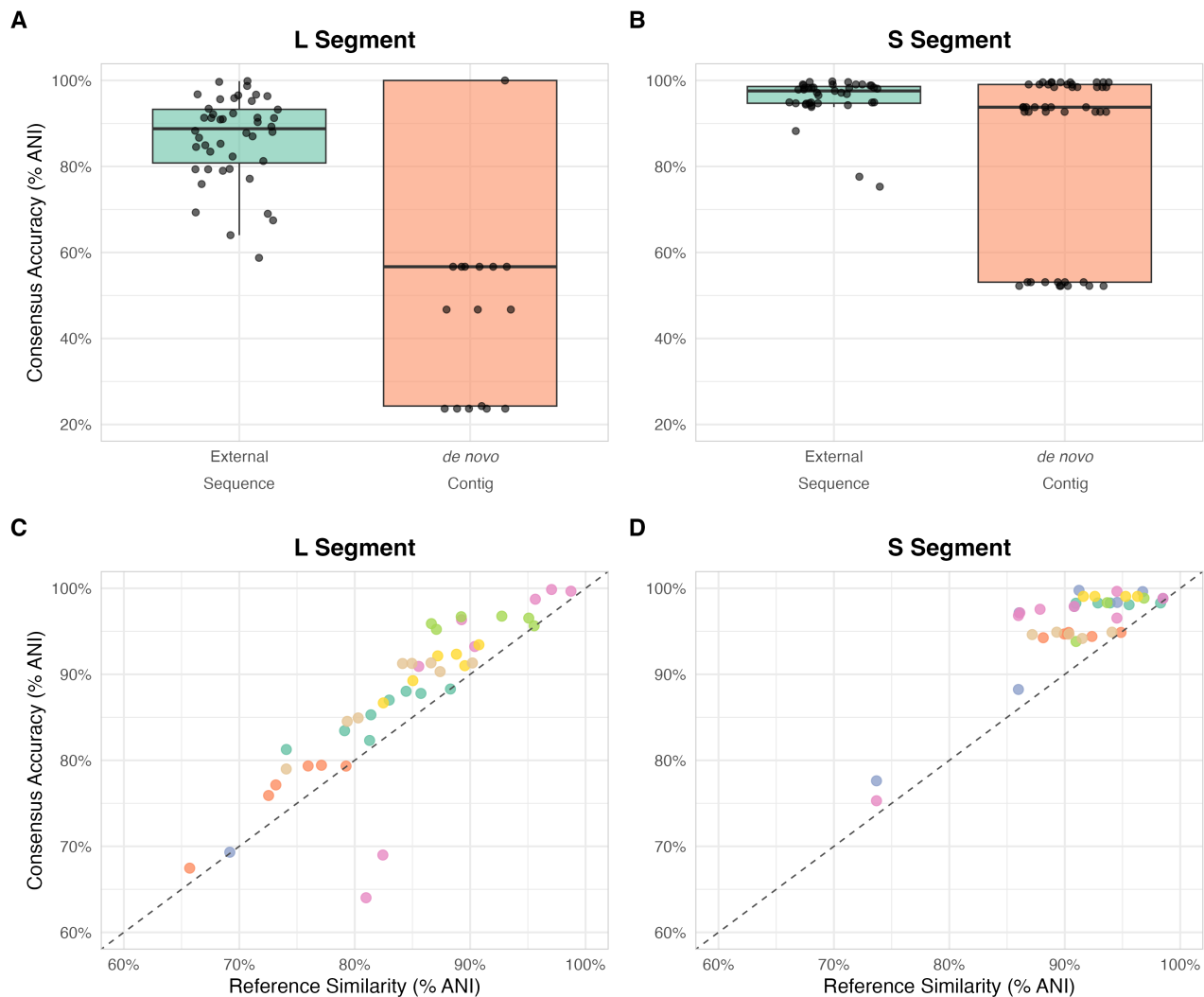


Fig. 2. Impact of scaffolding reference on consensus reconstruction relative to a U-RVDB baseline. (A, B) Consensus genome ANI (%) to baseline using external reference sequences vs. de novo contigs for L and S segments. (C, D) Correlation between scaffolding reference ANI and resulting consensus ANI to baseline for L and S segments, colored by sample.

Conflict of Interest

The authors declare no competing interests.

References

- S F Altschul, et al. Basic local alignment search tool. *J. Mol. Biol.*, 215:403–410, 1990. doi: 10.1016/S0022-2836(05)80360-2.
- Jasmijn A Baaijens, Amal Zine El Aabidine, Eric Rivals, and Alexander Schönhuth. De novo assembly of viral quasiespecies using overlap graphs. *Genome Res.*, 27:835–848, 2017. doi: 10.1101/gr.215038.116.
- Irene Bassano, et al. Evaluation of variant calling algorithms for wastewater-based epidemiology using mixed populations of SARS-CoV-2 variants in synthetic and wastewater samples. *bioRxiv*, 2022. doi: 10.1101/2022.06.06.22275866.
- Anthony M Bolger, Marc Lohse, and Bjoern Usadel. Trimmomatic: a flexible trimmer for illumina sequence data. *Bioinformatics*, 30:2114–2120, 2014. doi: 10.1093/bioinformatics/btu170.
- Brian Bushnell. BMap.
- Vito Adrian Cantu, Jeffrey Sadural, and Robert Edwards. PRINSEQ++, a multi-threaded tool for fast and efficient quality control and preprocessing of sequencing datasets. *PeerJ*, 2019. doi: 10.7287/peerj.preprints.27553v1.
- Shifu Chen, Yanqing Zhou, Yaru Chen, and Jia Gu. fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34:i884–i890, 2018. doi: 10.1093/bioinformatics/bty560.
- Petr Danecek, et al. Twelve years of SAMtools and BCFtools. *Gigascience*, 10, 2021. doi: 10.1093/gigascience/giab008.
- Jutte J C de Vries, et al. Benchmark of thirteen bioinformatic pipelines for metagenomic virus diagnostics using datasets from clinical samples. *J. Clin. Virol.*, 141:104908, 2021. doi: 10.1016/j.jcv.2021.104908.
- Paolo Di Tommaso, et al. Nextflow enables reproducible computational workflows. *Nat. Biotechnol.*, 35:316–319, 2017. doi: 10.1038/nbt.3820.

- Philip Ewels, Måns Magnusson, Sverker Lundin, and Max Källér. MultiQC: summarize analysis results for multiple tools and samples in a single report. *Bioinformatics*, 32: 3047–3048, 2016a. doi: 10.1093/bioinformatics/btw354.
- Philip A Ewels, et al. The nf-core framework for community-curated bioinformatics pipelines. *Nat. Biotechnol.*, 38:276–278, 2020b. doi: 10.1038/s41587-020-0439-x.
- Anne Gleizes, et al. Virosaurus a reference to explore and capture virus genetic diversity. *Viruses*, 12, 2020. doi: 10.3390/v12111248.
- Norman Goodacre, et al. A reference viral database (RVDB) to enhance bioinformatics analysis of high-throughput sequencing for novel virus detection. *mSphere*, 3, 2018. doi: 10.1128/mSphereDirect.00069-18.
- Manfred G Grabherr, et al. Full-length transcriptome assembly from RNA-seq data without a reference genome. *Nat. Biotechnol.*, 29:644–652, 2011. doi: 10.1038/nbt.1883.
- Nathan D Grubaugh, et al. An amplicon-based sequencing framework for accurately measuring intrahost virus diversity using PrimalSeq and iVar. *Genome Biol.*, 20:8, 2019. doi: 10.1186/s13059-018-1618-7.
- Jiarong Guo, et al. VirSorter2: a multi-classifier, expert-guided approach to detect diverse DNA and RNA viruses. *Microbiome*, 9:37, 2021. doi: 10.1186/s40168-020-00990-y.
- Kazutaka Katoh, Kazuharu Misawa, Kei-Ichi Kuma, and Takashi Miyata. MAFFT: a novel method for rapid multiple sequence alignment based on fast fourier transform. *Nucleic Acids Res.*, 30:3059–3066, 2002. doi: 10.1093/nar/gkf436.
- Kristopher Kieft, et al. vRhyme enables binning of viral genomes from metagenomes. *Nucleic Acids Res.*, 50:e83, 2022a. doi: 10.1093/nar/gkac341.
- Kristopher Kieft, Zhichao Zhou, and Karthik Anantharaman. VIBRANT: automated recovery, annotation and curation of microbial viruses, and evaluation of viral community function from genomic sequences. *Microbiome*, 8:90, 2020b. doi: 10.1186/s40168-020-00867-0.
- Gregory M Kurtzer, Vanessa Sochat, and Michael W Bauer. Singularity: Scientific containers for mobility of compute. *PLoS One*, 12:e0177459, 2017. doi: 10.1371/journal.pone.0177459.
- Björn E Langer, et al. Empowering bioinformatics communities with nextflow and nf-core. *Genome Biol.*, 26:228, 2025. doi: 10.1186/s13059-025-03673-9.
- Ben Langmead, Christopher Wilks, Valentin Antonescu, and Rone Charles. Scaling read aligners to hundreds of threads on general-purpose processors. *Bioinformatics*, 35:421–432, 2019. doi: 10.1093/bioinformatics/bty648.
- Jeroen F J Laros. HUMID: HUMID: reference free FastQ deduplication.
- Dinghua Li, et al. MEGAHIT v1.0: A fast and scalable metagenome assembler driven by advanced methodologies and community practices. *Methods*, 102:3–11, 2016. doi: 10.1016/j.ymeth.2016.02.020.
- Heng Li. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics*, 34:3094–3100, 2018. doi: 10.1093/bioinformatics/bty191.
- Weizhong Li and Adam Godzik. Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences. *Bioinformatics*, 22:1658–1659, 2006. doi: 10.1093/bioinformatics/btl158.
- Dmitry Meleshko, Iman Hajirasouliha, and Anton Korobeynikov. coronaSPAdes: from biosynthetic gene clusters to RNA viral assemblies. *Bioinformatics*, 2021. doi: 10.1093/bioinformatics/btab597.
- Peter Menzel, Kim Lee Ng, and Anders Krogh. Fast and sensitive taxonomic classification for metagenomics with kaiju. *Nat. Commun.*, 7:11257, 2016. doi: 10.1038/ncomms11257.
- D Merkel. Docker: lightweight linux containers for consistent development and deployment. *Linux Journal*, 2014:2, 2014. doi: 10.5555/2600239.2600241.
- Stephen Nayfach, et al. CheckV assesses the quality and completeness of metagenome-assembled viral genomes. *Nat. Biotechnol.*, 39:578–585, 2021. doi: 10.1038/s41587-020-00774-7.
- Brian D Ondov, Nicholas H Bergman, and Adam M Phillippy. Interactive metagenomic visualization in a web browser. *BMC Bioinformatics*, 12:385, 2011a. doi: 10.1186/1471-2105-12-385.
- Brian D Ondov, et al. Mash screen: high-throughput sequence containment estimation for genome discovery. *Genome Biol.*, 20:232, 2019b. doi: 10.1186/s13059-019-1841-x.
- Guillermo Rangel-Pineros, et al. VIRify: an integrated detection, annotation and taxonomic classification pipeline using virus-specific protein profile hidden markov models. *bioRxiv*, page 2022.08.22.504484, 2022. doi: 10.1101/2022.08.22.504484.
- Torbjørn Rognes, et al. VSEARCH: a versatile open source tool for metagenomics. *PeerJ*, 4:e2584, 2016. doi: 10.7717/peerj.2584.
- Samuel S Shepard, et al. Viral deep sequencing needs an adaptive approach: IRMA, the iterative refinement meta-assembler. *BMC Genomics*, 17:708, 2016. doi: 10.1186/s12864-016-3030-6.
- Tom Smith, Andreas Heger, and Ian Sudbery. UMI-tools: modeling sequencing errors in unique molecular identifiers to improve quantification accuracy. *Genome Res.*, 27:491–499, 2017. doi: 10.1101/gr.209601.116.
- Martin Steinegger and Johannes Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat. Biotechnol.*, 35:1026–1028, 2017. doi: 10.1038/nbt.3988.
- Md Vasmuddin, Sanchit Misra, Heng Li, and Srinivas Aluru. Efficient architecture-aware acceleration of BWA-MEM for multicore systems. In *2019 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 314–324. IEEE, 2019. doi: 10.1109/IPDPS.2019.00041.
- Derrick E Wood, Jennifer Lu, and Ben Langmead. Improved metagenomic analysis with kraken 2. *Genome Biol.*, 20:257, 2019. doi: 10.1186/s13059-019-1891-0.
- Andrzej Zielezinski, et al. Ultrafast and accurate sequence alignment and clustering of viral genomes. *Nat. Methods*, 22:1191–1194, 2025. doi: 10.1038/s41592-025-02701-7.